

**To Grad or Not to Grad?**  
**An Econometric Approach to Estimating the Return**  
**to Graduate Degrees Relative to Bachelor's Degrees**  
**Across Canada**

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## **Abstract**

This paper estimates how the wage premium for a graduate degree relative to a bachelor's degree varies across Canadian provinces. Using the 2021 Census Public Use Microdata File (n=127,765), I estimate a Mincer-style log wage equation with province fixed effects and interactions between Graduate Degree and Province, checking the results with a Tobit model that accounts for top and bottom coding of wages. Graduate degree holders earn an estimated 16.8% more than bachelor's degree holders in the pooled model, while the province estimates range from 7.7% in Alberta to 43.6% in Prince Edward Island. The premium is statistically distinguishable from zero in eight out of ten provinces. A joint test rejects equal premiums across provinces ( $F=4.49$ ,  $p<0.001$ ). The three Prairie provinces have a significantly lower average premium than Ontario (20.9%) and than the average of Prince Edward Island, Nova Scotia, and New Brunswick. Because graduate study is a choice, and the available instruments were too weak for two-stage least squares, all estimates are interpreted as correlations rather than causal effects.

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# 1 Introduction

## 1.1 Motivations

The decision to pursue graduate studies is perhaps the most significant financial and personal gamble a student will face in their early career. It’s a dilemma familiar to any student who has sat in an upper year lecture, surrounded by ambitious peers deliberating their futures. Some will immediately head to law school or an MBA program, while others dive straight into the workforce. A few brave ones commit to five years of doctoral study. But who made the right choice?

In 2020, roughly 226,971 Canadians earned a bachelor’s degree and contemplated whether to pursue graduate education (Statistics Canada, 2025). Students in this position instinctively weigh the trade-offs, but in the absence of a definite payoff, it becomes rather difficult to determine what is truly “worth it.” For instance, Graduate programs require substantial investments of time, money, and an opportunity cost of delayed career entry, however the return remains uncertain. Does a master’s degree justify these costs? Will PhD holders earn enough to compensate for their extended time out of the labour market? And do these answers vary depending on whether you plan to work in Ontario, Nova Scotia, or Alberta?

## 1.2 Contribution

Using microdata from Statistics Canada’s 2021 Census, this paper will estimate how returns to graduate education compare with a bachelor’s, with a particular focus on whether these returns are invariant across Canada, or a dynamic response to provincial labour market equilibria. These findings contribute to the Canadian returns to education literature by providing a provincial comparison of graduate degree returns which is underrepresented in recent research. From a policy perspective, my results inform prospective graduate stu-

dents enrolment decisions and suggest that provincial labour markets value credentials quite differently.

## 2 Literature Review

The foundation for analyzing returns to education lies in human capital theory. Developed by Gary Becker and Theodore Schultz, this theory recognizes education as an investment that improves worker productivity, suggesting that individuals attain higher education and incur the affiliated costs, with the expectation that future earnings will justify the investment (Eide & Showalter, 2010). Graduate education represents an application of this theory, as it requires opportunity and direct costs, with individuals choosing to enrol based on the expectation, subject to uncertainty, that the future benefits are valued higher than these costs.

To quantify this return, the Mincer earnings function has become the standard framework to estimate wage returns (Heckman et al., 2003). Jacob Mincer’s model relates log earnings to schooling and experience. He originally used 1960 U.S. Census data to find that each additional year of schooling raised earnings by approximately 10%. The model assumes linearity, in that every extra year of education raises earnings by the same percentage. (Clark & Nielsen, 2024). While seemingly restrictive, a 2014 study reported that the basic model is quite stable, validating its continued use as a benchmark (Montenegro & Patrinos, 2014).

Existing literature widely regards ability bias as a central identification challenge. OLS estimates tend to be biased upward, since high ability positively influences education and wage. This endogeneity problem has motivated instrumental variable approaches. For instance, David Card’s work used college proximity as instruments, and demonstrated that men who grew up near colleges with 4 year programs obtained more education and earned higher wages. His results showed that instead of correcting upward OLS bias, IV estimates were even higher (Card, 1993). Card cautioned that IV estimates identify average effects for

the group of individuals whose schooling decisions were influenced by the instrument, rather than identifying an average for the broader population. He also noted that estimates tend to be imprecise, as no instrument achieves true randomization (Card, 2001).

In the Canadian context, Canada’s labour market is defined by exceptionally high post-secondary attainment, largely due to the many college and trades options (Boothby & Drewes, 2006). Despite these diverse pathways, a persistent “skill premium” exists for university graduates. Census data has demonstrated that the wage differential for bachelor’s degrees increased steadily from 1980 to 2005 (Boudarbat, et al., 2010). However, these measured monetary returns likely represent a lower bound of the “true” return to schooling, as standard earnings functions typically omit non-monetary private benefits and social externalities (Cohn, 1997). That being said, Canadian educational investments have demonstrated modest to high returns once both market and non-market effects have been considered (Cohn, 1997).

## **3 Data**

### **3.1 Data Source**

This study uses the 2021 Canada Census of Population Public Use Microdata Individuals File, published by Statistics Canada, containing demographic, geographic, income, social, and education variables. Since this study specifically looks into the returns of higher education, I subset the sample to only include observations for completed bachelor’s and graduate degrees, excluding those in the trades, college, or non academic post-secondary pathways. This ensures a more comparable set to draw conclusions from, as these degrees follow a similar academic trajectory. For this reason, I also omit the three territories, due to different labour and education patterns. Non-available or non-applicable categories were assigned NA values. This leaves us with 127,765 observations.

## 3.2 Variable Selection

- **Dependent Variable:** For our dependent variable, we use the logarithm of wage. This is standard in Mincer equations for two primary reasons. Firstly, it allows for the coefficients to be interpreted as percentage changes, which is more economically meaningful when comparing wage returns. Secondly, the log transformation normalizes the distribution to address right-skewed wage data (Feng et al., 2014). The variable, *Wages*, refers to the gross wages and salaries before deductions, such as income taxes or pension plan contributions.
- **Independent Variables:** Our first independent variable of interest is *Graduate*, a binary variable taking a value of 1 if the individual has completed a graduate degree (includes Master’s and Doctoral degrees, as well as professional degrees in Law and Medicine), and 0 if their highest education is a completed bachelor’s degree. We are also interested in *Province*, a fixed effect variable, which includes 9 binary variables corresponding to each province, with Ontario set as the base.
- **Controls:** The remaining controls include *Age* (numerical, proxy for experience), *STEM* (science, technology, engineering, and math fields = 1, otherwise = 0), *Female* (female = 1, male = 0), *Immigrant* (immigrant = 1, Canadian born = 0), *NonPermRes* (non-permanent resident = 1, permanent resident = 0), *Indigenous* (Indigenous = 1, otherwise = 0), *LargeCity* (living in Toronto, Ottawa, Montreal, Calgary, Edmonton, or Vancouver = 1, otherwise = 0), and *FullTime* (employed in a full-time position = 1, otherwise = 0).

## 4 Methodology

The empirical method is motivated by Mincer specification, as discussed in the Literature Review. The model in this paper will build on the original by adding controls, province

fixed effects, and consider potential instruments to address endogeneity. Lastly, we will estimate a Tobit model due to censoring reported in the wage data. The results will be compared with the OLS estimates, to determine if the censoring has a significant effect on the estimators.

## 4.1 Ordinary Least Squares

Our primary model is defined as the following:

$$\log(wage_j) = \beta_0 + \beta_1 Graduate_j + \sum_{p=1}^P \delta_p Province_{jp} + \sum_{p=1}^P \gamma_p (Graduate_j \times Province_{jp}) + X_j' \beta + u_j$$

where  $Graduate_j$  is a binary variable for the attainment of a graduate degree,  $Province_{jp}$  represents province fixed effects,  $X_j$  is a vector of controls, and  $\gamma_p$  captures the interaction of graduate degree returns across provinces.

This model can be simplified to:

$$Y_j = X_j' \beta + u_j$$

The derivation of the OLS estimator  $\hat{\beta}$  is:

$$\hat{\beta} = \arg \min_{\beta} \sum_{j=1}^n (Y_j - X_j' \beta)^2$$

Taking the first order condition and solving, we arrive at:

$$\hat{\beta} = \left( \sum_{j=1}^n X_j X_j' \right)^{-1} \left( \sum_{j=1}^n X_j Y_j \right)$$

## Biasedness

To determine if the OLS estimator is biased, we take the expectation of  $\hat{\beta}$ :

$$E[\hat{\beta}] = E \left[ \left( \frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left( \frac{1}{n} \sum_{j=1}^n X_j Y_j \right) \right]$$
$$E[\hat{\beta}] = \beta + E \left[ \left( \frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left( \frac{1}{n} \sum_{j=1}^n X_j u_j \right) \right]$$

If  $E[u | X] = 0$ , the second term is zero by the law of iterated expectations and  $\hat{\beta}$  is unbiased.

If  $X$  is endogenous, then  $E[Xu] \neq 0$ , the second term does not vanish, and by the Law of Large Numbers:

$$\hat{\beta} \xrightarrow{p} \beta + (E[XX'])^{-1} E[Xu]$$

## Asymptotic Normality

One of the assumptions for OLS is homoskedasticity. Thus, a Breusch-Pagan test was run to determine how to construct appropriate standard errors. We strongly reject homoskedasticity ( $p < 0.001$ ), and report heteroskedasticity-robust standard errors.

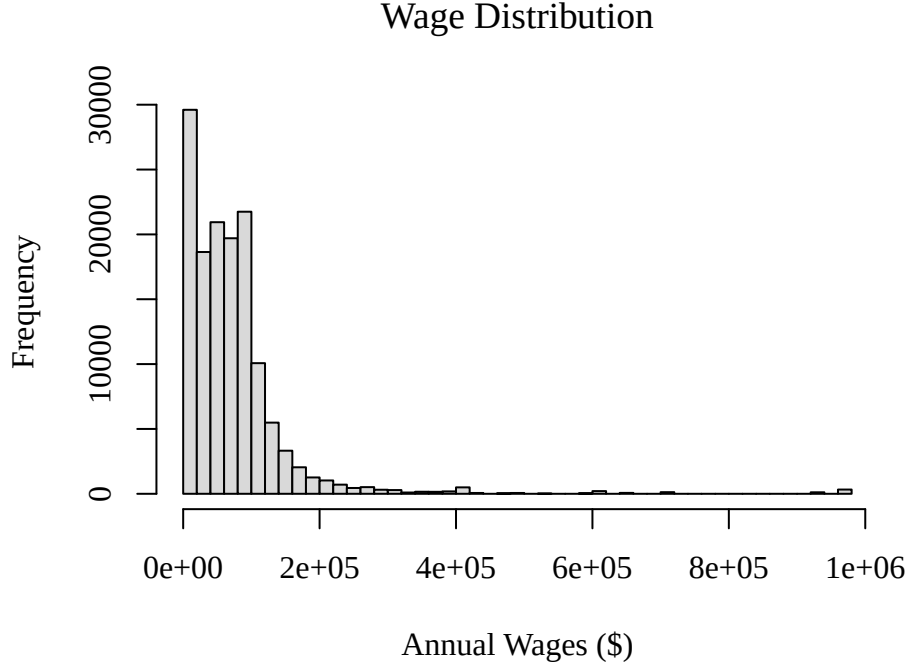
By the Law of Large Numbers and Central Limit Theorem, the asymptotic normality is:

$$\sqrt{n}(\hat{\beta} - \beta) = \left( \frac{1}{n} \sum_{j=1}^n X_j X_j' \right)^{-1} \left( \frac{1}{\sqrt{n}} \sum_{j=1}^n X_j u_j \right) \xrightarrow{d} \mathcal{N} \left( 0, (E[XX'])^{-1} E[XX'u^2] (E[XX'])^{-1} \right)$$

If  $X$  is endogenous,  $\beta$  differs from the true value by the bias term above, so the estimates are interpreted as correlations, and the robust standard errors estimate this variance.

## 4.2 Tobit Model Approach

The Census documentation noted that the variable *Wage* had instances of top coding and bottom coding. This distorts both ends of the distribution by treating them as having identical income values regardless of their true earnings. Specifically, the lower end is bottom coded at \$1 (1.7%) and the upper end, top coded at \$967,998 (0.3%). One way to address censoring is by Tobit estimation, an MLE method that treats the observed  $\log(\text{wage})$  as censored realizations of a latent variable  $Y_i^*$ . Although the proportion of censored observations is relatively small, the Tobit model is employed as a robustness check to ensure that the OLS estimates are not significantly biased by the restricted range of the dependent variable. Under the assumptions that the observations are i.i.d. and that the errors are normally distributed with constant variance conditional on the regressors, the Tobit maximum likelihood estimator is consistent for the coefficients on the latent wage. Because the Breusch-Pagan test rejects homoskedasticity in the OLS model, these assumptions may not hold exactly, so the Tobit model serves as a robustness check, and the closeness of its estimates to the OLS estimates is the relevant evidence. The following figure is a histogram of the wage distribution.



**Figure 1:** Distribution of Annual Wages

The distribution is right-skewed and exhibits censoring at both extremes. As shown in Figure 1, the impact of censoring is visually more prominent at the lower bound. Scientific notation on the x-axis represents wages in Canadian dollars (e.g., 2e+05 denotes \$200,000).

## Tobit Model Derivation

The latent wage,  $Y_j^*$ , represents the true earnings of an individual but is only observed within the range  $[B_L, B_U]$ . The latent variable follows the model:

$$Y_j^* = X_j' \beta + u_j, \quad u_j \mid X_j \sim \mathcal{N}(0, \sigma^2)$$

where,

$$Y_j = \begin{cases} B_L & \text{if } Y_j^* \leq B_L \\ Y_j^* & \text{if } B_L < Y_j^* < B_U \\ B_U & \text{if } Y_j^* \geq B_U \end{cases}$$

The log-likelihood is:

$$\begin{aligned} \ell(\beta, \sigma) = \frac{1}{n} \sum_{j=1}^n & \left[ \mathbf{1}(B_L < Y_j < B_U) \log \left( \frac{1}{\sigma} \phi \left( \frac{Y_j - X_j' \beta}{\sigma} \right) \right) \right. \\ & \left. + \mathbf{1}(Y_j = B_L) \log \left( \Phi \left( \frac{B_L - X_j' \beta}{\sigma} \right) \right) + \mathbf{1}(Y_j = B_U) \log \left( 1 - \Phi \left( \frac{B_U - X_j' \beta}{\sigma} \right) \right) \right] \end{aligned}$$

where  $\phi$  and  $\Phi$  denote the standard normal density and cumulative distribution function, respectively. Maximizing the log-likelihood with respect to  $\beta$  and  $\sigma$  yields the Tobit estimators.

### 4.3 Addressing the Endogeneity of Graduate

It can be argued quite easily that our education measurement, *Graduate*, is endogenous. One argument is that it is a choice variable. That is, individuals make their own decision on how much, and what type of, education to pursue, and often base this choice on unobserved influences. Many of these influences, notably ability, access, preferences, and family influence are captured in the error term, creating omitted variable bias. Additionally, individuals may be incentivized to acquire higher education, with the anticipation of higher future wages, creating a potential source of selection bias. Measurement errors present another source of endogeneity. Since census data is self-reported, there is a potential for misclassification. By way of illustration, graduate students who are working but have not yet earned their degree may be incorrectly categorized. This endogeneity prevents us from observing a causal relationship, as coefficients are biased (likely upwards) and inconsistent. All OLS regression

results will be interpreted as correlations.

Ideally, we would use two instruments to capture the part of *Graduate* that is not correlated with the error, however, two problems are encountered in this process. First, finding instruments that satisfy validity and relevance in census data is difficult. This is because most variables in the dataset that are correlated with education, affect wages directly. When we do find exogenous variables, they are often not correlated enough with education. Second, the interaction of *Graduate* and *Province* creates ten endogenous variables that must be instrumented for, instead of just one.

The two stage least squares estimation was attempted. An extensive search for suitable instruments within the 2021 Census PUMF was conducted, and the instruments selected were *Parental Place of Birth* (primary household parent born outside of Canada = 1, otherwise = 0) and *Second Generation* (second generation = 1, otherwise = 0). The Hausman test rejected the exogeneity of *Graduate* ( $p=0.003$ ), and the Sargan test failed to reject the exogeneity of the instruments ( $p=0.346$ ). The F-test reported the instruments were strong for *Graduate:ProvinceQC* ( $F=54.723$ ), *Graduate:ProvinceAB* ( $F=18.876$ ), *Graduate:ProvinceBC* ( $F=16.949$ ), and *Graduate:ProvinceNS* ( $F=10.100$ ), but weak for *Graduate* ( $F=5.057$ ), *Graduate:ProvincePEI* ( $F=1.503$ ), *Graduate:ProvinceNL* ( $F=3.888$ ), *Graduate:ProvinceSK* ( $F=6.585$ ), *Graduate:ProvinceMB* ( $F=3.751$ ), and *Graduate:ProvinceNB* ( $F=3.934$ ).

Despite the validity from the Sargan test, economic reasoning makes it possible that family background does not influence education enough. In fact, recent literature has cautioned the use of family background instruments as they are typically considered “weak or invalid” (Clark & Nielsen, 2024). Given the variables to choose from in census data, parental place of birth and second generation status had the strongest results as instrument contenders, but not strong enough to implement 2SLS inference.

## 5 Results

All regression estimates in this section are interpreted as correlations. I will use the  $(e^\beta - 1)$  transformation, report interactions as partial effects, and for all observations, hold the remaining regressors constant, where the vector of regressors is *Graduate*, *Province*, *STEM*, *Age*, *Female*, *LargeCity*, *FullTime*, *Immigrant*, *NonPermRes*, and *Indigenous*.

### 5.1 Model 1: Ordinary Least Squares

Table 1: OLS Estimates for Determinants of Wage

|             | Estimate | Robust SE | t value | p-value |
|-------------|----------|-----------|---------|---------|
| (Intercept) | 8.484    | 0.027     | 317.404 | 0.000   |
| Graduate    | 0.155    | 0.008     | 19.879  | 0.000   |
| STEM        | 0.092    | 0.008     | 11.800  | 0.000   |
| LargeCity   | 0.057    | 0.008     | 7.134   | 0.000   |
| ProvinceNL  | -0.096   | 0.039     | -2.439  | 0.015   |
| ProvincePEI | -0.059   | 0.037     | -1.587  | 0.113   |
| ProvinceNS  | -0.168   | 0.024     | -6.959  | 0.000   |
| ProvinceNB  | -0.094   | 0.025     | -3.852  | 0.000   |
| ProvinceQC  | -0.090   | 0.009     | -10.085 | 0.000   |
| ProvinceMB  | -0.036   | 0.018     | -1.970  | 0.049   |
| ProvinceSK  | -0.035   | 0.025     | -1.384  | 0.166   |
| ProvinceAB  | 0.048    | 0.011     | 4.406   | 0.000   |
| ProvinceBC  | -0.036   | 0.010     | -3.430  | 0.001   |
| Age         | 0.096    | 0.002     | 41.693  | 0.000   |
| Female      | -0.150   | 0.007     | -21.541 | 0.000   |
| Immigrant   | -0.283   | 0.008     | -36.989 | 0.000   |
| NonPermRes  | -0.590   | 0.019     | -31.148 | 0.000   |
| Indigenous  | -0.034   | 0.024     | -1.397  | 0.162   |
| FullTime    | 1.650    | 0.015     | 113.393 | 0.000   |
| I(Age^2)    | -0.001   | 0.000     | -32.950 | 0.000   |

**Note:** N = 127,765, R-squared = 0.2357, Adj. R-squared = 0.2356, F-statistic = 2,074 (p < 0.001)

Table 1 presents the OLS estimates of our basic model. The regression results indicate that when Canadians hold a graduate degree, they earn, on average, 16.8% higher wages

than those with a bachelor's ( $p < 0.001$ ). Individuals who studied in a STEM major earn, on average, 9.6% higher wages, than those in non-STEM disciplines ( $p < 0.001$ ). Individuals who live in a large city earn, on average, 5.9% higher wages than those who don't ( $p < 0.001$ ). The results demonstrate variation in wages between provinces, with most showing a lower wage return than in Ontario. Individuals in Alberta however, earn approximately 4.9% higher wages than those in Ontario ( $p < 0.001$ ). The marginal effect of *Age* demonstrates a positive, diminishing relationship, which is consistent with the existing literature ( $p < 0.001$ ). Wages increase at a diminishing rate until 47.3 years of age, after which it declines. Among all university degree holders, *Female* wages are, on average, 13.9% lower than males ( $p < 0.001$ ). *Immigrant* and *Non-Permanent Resident* status, both have negative partial effects, earning, on average, 24.6% ( $p < 0.001$ ) and 44.6% ( $p < 0.001$ ) less than non-immigrants and permanent residents. Canadians with *Indigenous* status also exhibit a negative partial effect, however it is not significant. *Full Time* employment is highly correlated with wage ( $p < 0.001$ ).

## 5.2 Model 2: Ordinary Least Squares With Interaction

Table 2: OLS Estimates for Returns to Education

|                      | Estimate | Robust SE | t value | p-value |
|----------------------|----------|-----------|---------|---------|
| (Intercept)          | 8.475    | 0.027     | 314.182 | 0.000   |
| Graduate             | 0.190    | 0.011     | 16.612  | 0.000   |
| STEM                 | 0.092    | 0.008     | 11.814  | 0.000   |
| LargeCity            | 0.057    | 0.008     | 7.105   | 0.000   |
| ProvinceNL           | -0.078   | 0.043     | -1.798  | 0.072   |
| ProvincePEI          | -0.103   | 0.045     | -2.302  | 0.021   |
| ProvinceNS           | -0.175   | 0.027     | -6.475  | 0.000   |
| ProvinceNB           | -0.111   | 0.028     | -3.950  | 0.000   |
| ProvinceQC           | -0.072   | 0.010     | -6.961  | 0.000   |
| ProvinceMB           | -0.014   | 0.020     | -0.681  | 0.496   |
| ProvinceSK           | -0.009   | 0.028     | -0.308  | 0.758   |
| ProvinceAB           | 0.078    | 0.012     | 6.459   | 0.000   |
| ProvinceBC           | -0.022   | 0.012     | -1.867  | 0.062   |
| Age                  | 0.096    | 0.002     | 41.638  | 0.000   |
| Female               | -0.151   | 0.007     | -21.557 | 0.000   |
| Immigrant            | -0.283   | 0.008     | -37.025 | 0.000   |
| NonPermRes           | -0.589   | 0.019     | -31.048 | 0.000   |
| Indigenous           | -0.035   | 0.024     | -1.436  | 0.151   |
| FullTime             | 1.650    | 0.015     | 113.421 | 0.000   |
| I(Age^2)             | -0.001   | 0.000     | -32.927 | 0.000   |
| Graduate:ProvinceNL  | -0.061   | 0.092     | -0.665  | 0.506   |
| Graduate:ProvincePEI | 0.173    | 0.076     | 2.280   | 0.023   |
| Graduate:ProvinceNS  | 0.027    | 0.057     | 0.474   | 0.635   |
| Graduate:ProvinceNB  | 0.082    | 0.055     | 1.509   | 0.131   |
| Graduate:ProvinceQC  | -0.061   | 0.020     | -3.083  | 0.002   |
| Graduate:ProvinceMB  | -0.094   | 0.046     | -2.056  | 0.040   |
| Graduate:ProvinceSK  | -0.108   | 0.064     | -1.680  | 0.093   |
| Graduate:ProvinceAB  | -0.115   | 0.027     | -4.335  | 0.000   |
| Graduate:ProvinceBC  | -0.047   | 0.024     | -1.950  | 0.051   |

**Note:** N = 127,765, R-squared = 0.236, Adj. R-squared = 0.2358, F-statistic = 1,409 (p < 0.001)

Table 3: Estimated Graduate Wage Premium (Model 2)

|    | Premium (%) | 95% CI lower | 95% CI upper | p-value |
|----|-------------|--------------|--------------|---------|
| ON | 20.878      | 18.204       | 23.612       | 0.000   |

|     | Premium (%) | 95% CI lower | 95% CI upper | p-value |
|-----|-------------|--------------|--------------|---------|
| NL  | 13.680      | -5.007       | 36.044       | 0.162   |
| PEI | 43.649      | 24.059       | 66.332       | 0.000   |
| NS  | 24.211      | 11.254       | 38.676       | 0.000   |
| NB  | 31.246      | 18.219       | 45.709       | 0.000   |
| QC  | 13.677      | 10.070       | 17.402       | 0.000   |
| MB  | 10.027      | 0.872        | 20.012       | 0.031   |
| SK  | 8.555       | -4.056       | 22.824       | 0.193   |
| AB  | 7.703       | 2.727        | 12.919       | 0.002   |
| BC  | 15.330      | 10.635       | 20.224       | 0.000   |

**Note:** The p-values test each premium against zero. Robust (HC3) standard errors.

Table 2 presents the OLS results of our primary model of focus. It differs from the previous as it includes the interaction between graduate and province, which is critical for observing heterogeneity between provinces. Based on the regression results, Canadians living in Ontario who hold a graduate degree, earn on average, 20.9% higher wages than those with a bachelor's ( $p < 0.001$ ). The results demonstrate heterogeneity between provinces, of individuals wages when they have a graduate degree, and a joint test rejects equal premiums across provinces  $F(9, 127,736) = (4.49, p < 0.001)$ . In the Atlantic provinces, the average marginal effects are 13.7% ( $p = 0.162$ ) in *Newfoundland & Labrador*, 43.6% ( $p < 0.001$ ) in *Prince Edward Island*, 24.2% ( $p < 0.001$ ) in *Nova Scotia*, and 31.2% ( $p < 0.001$ ) in *New Brunswick*. Moving inward, these partial effects are 13.7% ( $p < 0.001$ ) in *Quebec*, 10.0% ( $p = 0.031$ ) in *Manitoba*, and 8.6% in *Saskatchewan* ( $p = 0.193$ ). Finally, out west, we observe partial effects of 7.7% ( $p = 0.002$ ) in *Alberta* and 15.3% ( $p < 0.001$ ) in *British Columbia*. These p-values test each province's premium against zero, and the full estimates with confidence intervals are in Table 3. The interaction terms in Table 2 instead test whether a province's premium differs from Ontario's.

### 5.3 Model 3: Tobit

Table 4: Tobit Estimates for Returns to Education

|                      | Estimate | Robust SE | t value | p-value |
|----------------------|----------|-----------|---------|---------|
| (Intercept)          | 8.469    | 0.027     | 311.162 | 0.000   |
| Graduate             | 0.192    | 0.012     | 16.585  | 0.000   |
| STEM                 | 0.091    | 0.008     | 11.655  | 0.000   |
| LargeCity            | 0.059    | 0.008     | 7.317   | 0.000   |
| ProvinceNL           | -0.079   | 0.044     | -1.812  | 0.070   |
| ProvincePEI          | -0.103   | 0.045     | -2.311  | 0.021   |
| ProvinceNS           | -0.176   | 0.027     | -6.473  | 0.000   |
| ProvinceNB           | -0.112   | 0.028     | -3.959  | 0.000   |
| ProvinceQC           | -0.074   | 0.010     | -7.108  | 0.000   |
| ProvinceMB           | -0.014   | 0.020     | -0.675  | 0.500   |
| ProvinceSK           | -0.009   | 0.028     | -0.338  | 0.736   |
| ProvinceAB           | 0.076    | 0.012     | 6.283   | 0.000   |
| ProvinceBC           | -0.024   | 0.012     | -1.994  | 0.046   |
| Age                  | 0.097    | 0.002     | 41.456  | 0.000   |
| Female               | -0.153   | 0.007     | -21.690 | 0.000   |
| Immigrant            | -0.285   | 0.008     | -36.919 | 0.000   |
| NonPermRes           | -0.590   | 0.019     | -30.797 | 0.000   |
| Indigenous           | -0.035   | 0.024     | -1.448  | 0.148   |
| FullTime             | 1.654    | 0.015     | 112.415 | 0.000   |
| I(Age^2)             | -0.001   | 0.000     | -32.814 | 0.000   |
| Graduate:ProvinceNL  | -0.065   | 0.093     | -0.695  | 0.487   |
| Graduate:ProvincePEI | 0.170    | 0.075     | 2.266   | 0.023   |
| Graduate:ProvinceNS  | 0.025    | 0.058     | 0.434   | 0.664   |
| Graduate:ProvinceNB  | 0.080    | 0.055     | 1.470   | 0.141   |
| Graduate:ProvinceQC  | -0.064   | 0.020     | -3.174  | 0.002   |
| Graduate:ProvinceMB  | -0.097   | 0.046     | -2.103  | 0.036   |
| Graduate:ProvinceSK  | -0.110   | 0.064     | -1.711  | 0.087   |
| Graduate:ProvinceAB  | -0.118   | 0.027     | -4.413  | 0.000   |
| Graduate:ProvinceBC  | -0.049   | 0.024     | -2.039  | 0.041   |
| Log(scale)           | 0.178    | 0.008     | 22.036  | 0.000   |

**Note:** Log-Likelihood: -204,715.7, Scale Parameter ( $\sigma$ ): 1.195, N: 127,765

Table 5: Estimated Graduate Wage Premium (Model 3)

|     | Premium (%) | 95% CI lower | 95% CI upper | p-value |
|-----|-------------|--------------|--------------|---------|
| ON  | 21.115      | 18.404       | 23.888       | 0.000   |
| NL  | 13.542      | -5.217       | 36.014       | 0.168   |
| PEI | 43.598      | 24.152       | 66.091       | 0.000   |
| NS  | 24.187      | 11.155       | 38.748       | 0.000   |
| NB  | 31.252      | 18.199       | 45.746       | 0.000   |
| QC  | 13.639      | 10.013       | 17.385       | 0.000   |
| MB  | 9.970       | 0.795        | 19.979       | 0.033   |
| SK  | 8.474       | -4.206       | 22.832       | 0.200   |
| AB  | 7.608       | 2.612        | 12.847       | 0.003   |
| BC  | 15.267      | 10.548       | 20.187       | 0.000   |

**Note:** The p-values test each premium against zero. Robust (sandwich) standard errors.

Table 4 presents the MLE Tobit estimators. Based on this method, Canadians living in Ontario who hold a graduate degree, earn on average, 21.1% higher wages than those with a bachelor's ( $p < 0.001$ ). In the Atlantic provinces, the partial effects of graduate education on wage is 13.5% ( $p = 0.168$ ) in *Newfoundland & Labrador*, 43.6% ( $p < 0.001$ ) in *Prince Edward Island*, 24.2% ( $p < 0.001$ ) in *Nova Scotia*, and 31.3% ( $p < 0.001$ ) in *New Brunswick*. Moving inward, the partial effects are 13.6% ( $p < 0.001$ ) in *Quebec*, 9.97% ( $p = 0.033$ ) in *Manitoba*, and 8.5% ( $p = 0.200$ ) in *Saskatchewan*. Finally, out west, we observe partial effects of 7.6% ( $p = 0.003$ ) in *Alberta* and 15.3% ( $p < 0.001$ ) in *British Columbia*. These p-values test each province's premium against zero, and the full estimates with confidence intervals are in Table 5. The interaction terms in Table 4 instead test whether a province's premium differs from Ontario's. The convergence of the OLS and Tobit results despite the presence of top and bottom coding suggests that our estimates are not significantly influenced, with the Tobit premiums within 0.3 percentage points of the OLS premiums in every province.

## 6 Discussion

### 6.1 Economic Interpretation of Marginal Effects

Our results demonstrate a pattern of positive wage returns to graduate degrees relative to bachelor's degrees in Canada. The baseline OLS reported a pooled wage return that is about 16.8% higher than the return of a bachelor's degree, and for STEM majors, an additional return about 9.6% higher than non-STEM majors. The return to higher education aligns with human capital theory, in that the labour market tends to reward the skills and innovation associated with advanced academic qualification.

The second model, which interacted *Graduate* with *Province*, is more related to the focus of this study, which is to estimate differences in graduate wages across geographic regions. In particular, the return associated with graduate degrees in Ontario (20.9%) was significantly higher than in Quebec, Manitoba, and Alberta, which may be reflected by its labour market composition. Ontario is Canada's most populated province, with over 90% of its population concentrated in the south. Widely regarded as Canada's "economic hub," southern Ontario is dominated by finance, insurance, consulting, law, and professional sectors (CanadaVisa, 2026). Oftentimes, these roles require a master's degree, anywhere from an MBA, JD, or a specialized master's, providing a plausible case for why the wage premium seems more systematic than in other labour markets. This is consistent with economic signaling theory, which suggests that in the presence of asymmetric information, employers utilize graduate degrees as "signals" to identify capable candidates (Science Direct, n.d.).

Alberta's estimates present the most striking evidence of heterogeneity in this sample. Alberta is the only province where bachelor's degree holders earn more than in Ontario (8.0% higher), yet it has the lowest estimated premium for graduate education (7.7%), significantly below Ontario's ( $p < 0.001$  for the difference). This paradox is consistent with several structural characteristics of the provincial economy. First, Alberta consistently re-

ports the highest household median after-tax income in Canada (\$88,500 in 2023), as well as the highest real GDP per person (~\$72,000 in 2023) (Statistics Canada, 2023). Its renowned energy and natural resources industries are among the most productive sectors in Canada, suggesting that the marginal product of labour is higher in Alberta than in other provinces (Finlayson, 2026). This may also explain why skilled trades and field experience can outweigh academic qualification, keeping in mind that graduate degrees are still rewarded, but openings are not as common. A higher opportunity cost of pursuing graduate education is therefore very possible, when the alternative is so lucrative. This effect might extend to Alberta’s neighbouring Prairies, Manitoba and Saskatchewan, which rely heavily on agriculture and resource extraction. However, the resource sectors of these provinces don’t reach the same level of intensity as Alberta. This is reflected in slightly higher estimated premiums in Manitoba (10.0%) and Saskatchewan (8.6%) than in Alberta, although the intervals overlap substantially and Saskatchewan’s includes zero. Taken together, the three Prairie provinces have a significantly lower average premium than Ontario ( $F=13.1$ ,  $p<0.001$ ).

Quebec’s graduate premium (13.7%) falls in between Ontario and Alberta, and is significantly below Ontario’s ( $p=0.002$  for the difference), likely reflecting differences in labour market institutions. Quebec’s labour market is in many ways, unique within Canada, having the highest unionization rate (40% of all payroll employees in 2024), strong labour protections, and strict French language requirements enforced by Bill 101 (Note: this dataset was collected before Bill 96 was passed) (Institut de la statistique du Québec, 2025). The 1997 Canada-Quebec Labour Market Agreement in particular, transferred significant authority of employment management and training to the province. In combination with Quebec’s Pay Equity Act of 1996, these policies established requirements for pay equity within organizations, which may contribute to a less dispersed wage distribution (Government of Canada, 2020). Quebec’s language laws are also a defining feature, reducing accessibility for non-Francophone individuals. This structure is consistent with a reward for higher education, but with smaller wage gaps between education levels. The immigrant penalty (-24.6%) re-

ported in Table 1 is estimated across all provinces, so this paper cannot say whether barriers to labour market entry in Quebec contribute to it.

British Columbia’s graduate premium (15.3%) falls between Ontario’s and the Prairies’, and its difference from Ontario is borderline ( $p=0.051$ ). Its labour structure is often described as a “dual economy”, such that the northern is resource-based and the southern is technology-based (Caddell, 2018). According to Statistics Canada’s 2019 Labour Force Survey, occupations in construction, manufacturing, trade, professional services, educational services, and healthcare make up the majority of their labour force (Government of Canada, 2019). Thus, there exists a mix of sectors, some rewarding academics, and others, field experience where a bachelor’s degree is sufficient. This provides a plausible explanation of why the graduate premium remains significantly higher than Alberta’s ( $F=4.6$ ,  $p=0.033$ ), although this comparison was chosen after inspecting the estimates.

The Atlantic provinces show the widest range of estimates. Prince Edward Island had the largest estimated return at 43.6%, significantly above Ontario’s ( $p=0.023$  for the difference). However, its confidence interval (24.1% to 66.3%) is nearly eight times as wide as Ontario’s. As Canada’s least populated province, Prince Edward Island contributes only 98 graduate degree holders to the sample, compared with between 360 and 779 in the other Atlantic provinces and 16,059 in Ontario, which likely explains the imprecision of this estimate. Nova Scotia (24.2%) and New Brunswick (31.3%) also have large estimates with wide intervals, and neither can be distinguished from Ontario’s premium ( $p=0.635$  and  $p=0.131$ ). Newfoundland and Labrador’s estimate (13.7%) is closer to Quebec’s, but its interval includes zero. Three Atlantic provinces have higher point estimates than Ontario, although only Prince Edward Island’s difference is significant. As a group, Prince Edward Island, Nova Scotia, and New Brunswick have a significantly higher average premium than the Prairie provinces ( $F=19.8$ ,  $p<0.001$ ). That said, New Brunswick and Nova Scotia have placed a focus on marine and renewable innovation while facing an aging population, which is theoretically consistent with a demand for graduate degrees and a positive wage correlation,

however these estimates are imprecise, and this paper cannot test whether sectoral demand explains them (SR&ED, 2025).

From an economic theory perspective, the variation in returns across provinces suggests that the wage premium for graduate education may depend on the complementarity between the degree and the location. In this context, the lower returns in Alberta, and to a lesser extent Manitoba and Saskatchewan, are consistent with an environment where degrees and experience act more as substitutes, whereas the higher premium in Ontario can be interpreted as reflecting a setting where advanced education and experience are stronger complements. This paper does not observe occupations or industries, so it cannot test this account, and other explanations would produce the same pattern, including differences across provinces in who pursues graduate study, migration of degree holders, and field of study composition. The value of a graduate degree may therefore depend on the specific labour market in which it is utilized, though the estimates here do not identify that mechanism.

## 6.2 Limitations and Future Improvements

Several limitations constrain the scope of these findings, with the most significant being the inability to claim causal identification. Since *Graduate* is a choice variable, the model is subject to selection bias, and therefore endogeneity. The available instrumental variables in this dataset were too weak, preventing the 2SLS method from producing something causal. It is possible that the OLS estimates overstate the causal effect as choice motivators like ability, among others, cannot be controlled for.

A second important limitation is that wages are observed at a single point in time, such that measurements of how graduate degree returns evolve over a given career, are not captured by cross-sectional data. Future research with panel data could trace the earnings pattern for degree holders, to determine how the estimates differ for longer horizons.

Moreover, this study uses 2021 Census data, which was taken in the 2020 reference year,

a period that was irregularly affected by the COVID-19 pandemic. For many degree holders, the pandemic generated shocks, such as unemployment, layoffs, emergency benefits (CERB), and other distortions which are not representative of “typical” labour market conditions.

### 6.3 Conclusion

This paper set out to answer a deceptively simple question. Does a graduate degree pay off in Canada, and does the answer depend on the province in which you reside? The estimates indicate a positive premium for graduate education, though the difference is statistically distinguishable from zero in eight of ten, with its size varying across provinces.

The OLS and Tobit estimates consistently indicate a positive association between holding a graduate degree and wages relative to bachelor’s degree holders, with an estimated premium of 20.9% in Ontario. The more interesting story, however, is geographic. The interaction model shows that the estimated premium varies across provinces, and a joint test rejects equal premiums ( $F=4.49$ ,  $p<0.001$ ). The Prairie provinces have a significantly lower average premium than Ontario, and than the average of Prince Edward Island, Nova Scotia, and New Brunswick. Alberta has the lowest point estimate (7.7%), though its interval overlaps those of Manitoba and Saskatchewan.

The practical takeaway is that geographic location is correlated with graduate degree returns. Graduate study is a choice, and the available instruments were too weak to support a causal estimate, so unobserved ability and preferences may contribute to upward bias in the estimated premiums. Prospective students should consider their intended labour market when evaluating whether graduate education justifies its costs, recognizing estimates as a description of how wages differ between degree holders across provinces, rather than guaranteed outcomes.

## 7 Appendix

Table A1: TSLS Estimates for Returns to Education

|                      | Estimate | Robust.SE | t.value | p.value |
|----------------------|----------|-----------|---------|---------|
| (Intercept)          | 8.537    | 0.059     | 144.321 | 0.000   |
| Graduate             | 0.912    | 0.391     | 2.335   | 0.020   |
| STEM                 | 0.099    | 0.009     | 11.411  | 0.000   |
| LargeCity            | 0.043    | 0.012     | 3.522   | 0.000   |
| ProvinceNL           | -0.045   | 0.181     | -0.246  | 0.806   |
| ProvincePEI          | -0.254   | 0.308     | -0.825  | 0.410   |
| ProvinceNS           | -0.077   | 0.101     | -0.756  | 0.449   |
| ProvinceNB           | 0.110    | 0.134     | 0.820   | 0.412   |
| ProvinceQC           | 0.024    | 0.045     | 0.526   | 0.599   |
| ProvinceMB           | 0.375    | 0.263     | 1.426   | 0.154   |
| ProvinceSK           | 0.096    | 0.101     | 0.944   | 0.345   |
| ProvinceAB           | 0.164    | 0.059     | 2.796   | 0.005   |
| ProvinceBC           | -0.323   | 0.139     | -2.329  | 0.020   |
| Age                  | 0.075    | 0.012     | 6.227   | 0.000   |
| Female               | -0.132   | 0.013     | -9.863  | 0.000   |
| Immigrant            | -0.353   | 0.042     | -8.488  | 0.000   |
| NonPermRes           | -0.703   | 0.071     | -9.920  | 0.000   |
| Indigenous           | -0.035   | 0.029     | -1.186  | 0.236   |
| FullTime             | 1.632    | 0.021     | 77.091  | 0.000   |
| I(Age^2)             | -0.001   | 0.000     | -6.792  | 0.000   |
| Graduate:ProvinceNL  | -0.307   | 0.628     | -0.489  | 0.625   |
| Graduate:ProvincePEI | 0.715    | 1.183     | 0.604   | 0.546   |
| Graduate:ProvinceNS  | -0.344   | 0.398     | -0.864  | 0.388   |
| Graduate:ProvinceNB  | -0.866   | 0.606     | -1.428  | 0.153   |
| Graduate:ProvinceQC  | -0.442   | 0.168     | -2.636  | 0.008   |
| Graduate:ProvinceMB  | -1.735   | 1.295     | -1.340  | 0.180   |
| Graduate:ProvinceSK  | -0.459   | 0.411     | -1.115  | 0.265   |
| Graduate:ProvinceAB  | -0.330   | 0.239     | -1.379  | 0.168   |
| Graduate:ProvinceBC  | 1.055    | 0.501     | 2.103   | 0.035   |

Table A2: Sample Size by Province and Degree Level

|       | Bachelor's | Graduate | Total   | Graduate share (%) |
|-------|------------|----------|---------|--------------------|
| ON    | 39,854     | 16,059   | 55,913  | 28.7               |
| NL    | 807        | 360      | 1,167   | 30.8               |
| PEI   | 278        | 98       | 376     | 26.1               |
| NS    | 2,251      | 779      | 3,030   | 25.7               |
| NB    | 1,455      | 428      | 1,883   | 22.7               |
| QC    | 18,127     | 7,987    | 26,114  | 30.6               |
| MB    | 2,938      | 792      | 3,730   | 21.2               |
| SK    | 2,169      | 645      | 2,814   | 22.9               |
| AB    | 10,943     | 3,569    | 14,512  | 24.6               |
| BC    | 13,073     | 5,153    | 18,226  | 28.3               |
| Total | 91,895     | 35,870   | 127,765 | 28.1               |

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